

Auditing LLM Topic Labels with Unsupervised Text Clustering

Canadian Parliamentary Speeches, 1920–
1949

Your Name

Evaluating Econometric Methods | June 16, 2026

The Core Argument

LLM labels and KMeans clusters should not be treated as competing answers to the same question.

LLM LABELS

- ▶ classify speeches by **policy content**;
- ▶ provide broad, interpretable topic categories;
- ▶ serve as provisional semantic reference labels.

KMEANS CLUSTERS

- ▶ group speeches by **lexical similarity**;
- ▶ reveal repeated forms of parliamentary language;
- ▶ expose discourse structures inside and across topics.

Reframing the Question

INITIAL QUESTION

Is KMeans less accurate than LLM topic labeling?

This framing is too narrow, because it assumes that both methods are trying to recover the same object.

REVISED QUESTION

What does KMeans reveal that LLM topic labels do not show?

This framing turns mismatches into evidence about the structure of political language.

Final Research Question

When LLM-generated labels classify parliamentary speeches by policy content, what additional information can KMeans clusters provide about the discourse form of those speeches?

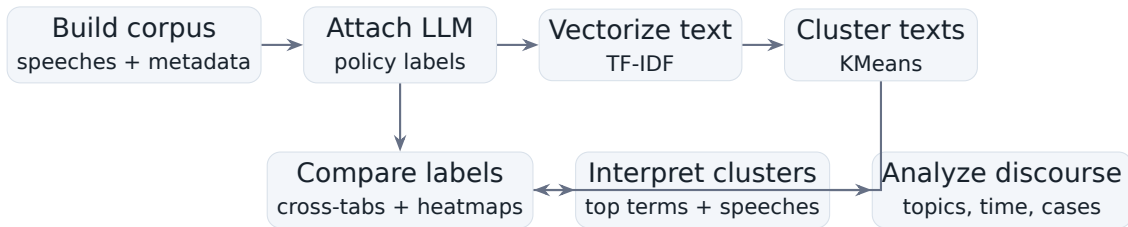
- ▶ The LLM label answers: **what policy topic is being discussed?**
- ▶ The KMeans cluster answers: **what kind of language is used to discuss it?**
- ▶ The main empirical object is therefore not simple agreement or disagreement, but the pattern of **structured mismatch**.

Data and Key Variables

Variable	Role in the analysis
speechtext	cleaned speech text used for modeling
speechtext_original	original text used for close reading and interpretation
year	year of speech, used for period and trend analysis
topic	original parliamentary topic description
category	LLM-generated broad policy topic label
kmeans_cluster	unsupervised cluster assignment from TF-IDF + KMeans
cluster_type	researcher-interpreted discourse type based on cluster meaning

Final full-corpus output: approximately 161,633 matched speeches from Canadian parliamentary debates, 1920–1949.

Research Workflow



The workflow treats LLM labels as useful but auditable, not as unquestioned ground truth.

Text Representation and Clustering

MODELING CHOICE

- ▶ Cleaned speeches are represented with TF-IDF features.
- ▶ KMeans is estimated with 20 clusters.
- ▶ The clustering is unsupervised: it does not use the LLM labels as training targets.

INTERPRETATION STRATEGY

- ▶ inspect top TF-IDF terms;
- ▶ compare cluster-category distributions;
- ▶ read representative original speeches;
- ▶ assign historically meaningful discourse interpretations.

The model is not designed to replace topic labels; it is designed to expose recurring patterns of language use.

How the Two Labeling Systems Are Compared

CATEGORY TO CLUSTER

- ▶ Start from an LLM policy category.
- ▶ Ask whether its speeches concentrate in one cluster or spread across several clusters.
- ▶ This tests whether a broad topic is linguistically homogeneous.

CLUSTER TO CATEGORY

- ▶ Start from a KMeans cluster.
- ▶ Ask whether it corresponds to one policy field or cuts across several fields.
- ▶ This identifies topic-specific clusters and cross-topic discourse forms.

The most informative cases are not only high-alignment clusters, but also systematic mismatches.

Finding 1: Some Clusters Align Clearly with Policy Topics

Cluster	Characteristic vocabulary	Dominant LLM category
19	tariff, duty, import, protection	International Trade and Tariffs
9	railway, line, Canadian National	Transportation and Communications
4	tax, income, revenue, taxation	Public Finance and Taxation
17	wheat, grain, bushel, wheat board	Agriculture and Food Policy
2	bank, loan, credit, money	Banking and Monetary Policy

Alignment is strongest when a policy field has distinctive, repeated, technical vocabulary.

Finding 2: A Single LLM Category Can Contain Multiple Discourse Forms

EXAMPLE: AGRICULTURE AND FOOD POLICY

- ▶ wheat and grain policy;
- ▶ farm prices and agricultural markets;
- ▶ inspection and product regulation;
- ▶ export rules;
- ▶ legal authority;
- ▶ federal-provincial administration;
- ▶ general parliamentary debate.

INTERPRETATION

The LLM category is semantically reasonable, but it hides the internal language structure of the topic.

KMeans separates different ways of speaking within the same policy area.

Finding 3: Some Clusters Capture Cross-Topic Discourse

Cluster	Interpretation	Why it crosses topics
15	general parliamentary discourse	common debate language across policy fields
5	legal and legislative language	statutes, authority, clauses, regulations
7	budgetary and expenditure language	appropriations, estimates, administrative costs
8	committee and procedural language	motions, amendments, committee-stage discussion
18	intergovernmental language	federal-provincial coordination and jurisdiction

A low match rate is not automatically a model failure; it may reveal a reusable parliamentary discourse form.

Close Reading: Same Topic, Different Forms

Speech ID	Shared policy topic	KMeans interpretation
570680	Inspection and Sale Act regulatory powers	legal and administrative authority
749345	amendment of dairy produce export regulations	clause-level amendment procedure

SPEECH 570680	SPEECH 749345
▶ Parliament ▶ authority ▶ jurisdiction ▶ minister and officials ▶ regulations	▶ simpler amendment ▶ strike out words ▶ paragraph (g) ▶ read this again ▶ procedural revision

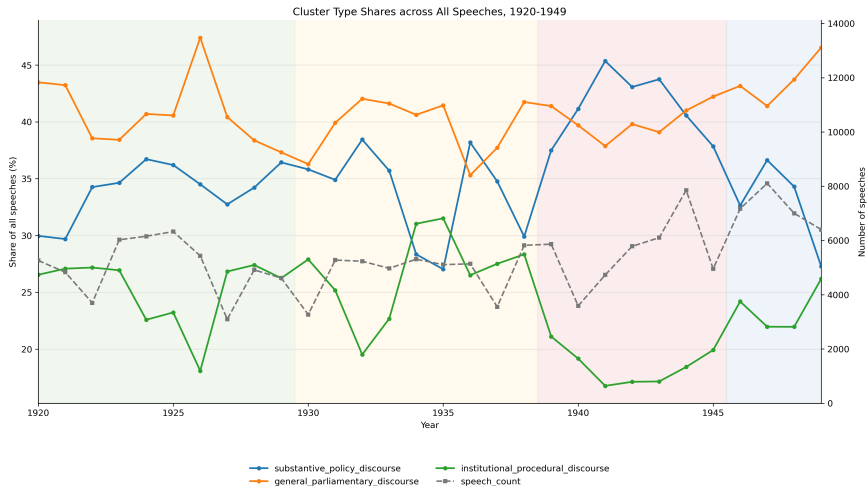
Same policy topic; different linguistic organization.

From 20 Clusters to Three Discourse Types

Discourse type	Clusters	Meaning
Substantive policy discourse	0, 1, 2, 4, 6, 9, 10, 11, 13, 14, 16, 17, 19	concrete policy content and sector-specific vocabulary
Institutional-procedural discourse	5, 7, 8, 18	legal, budgetary, committee, administrative, and intergovernmental language
General parliamentary discourse	3, 12, 15	broad debate, party politics, government-opposition rhetoric

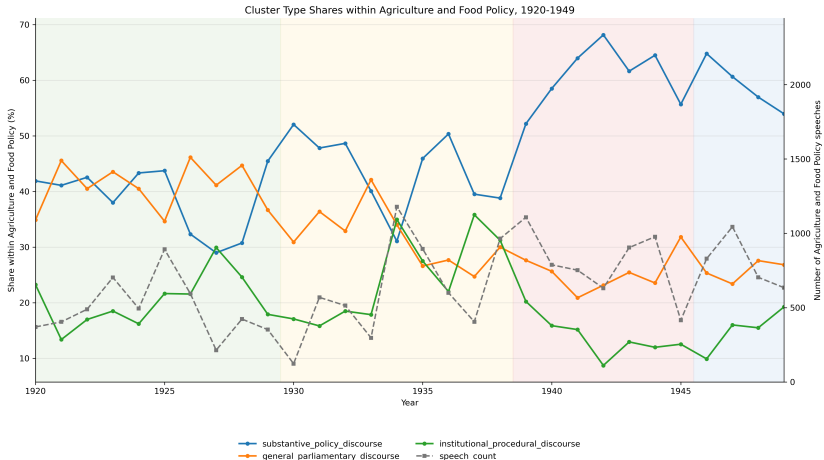
This grouping is interpretive. It is designed to make the cluster results historically readable, not to claim that the three types are automatically discovered ground truth.

Corpus-Level Pattern



The descriptive trend suggests that substantive policy discourse becomes especially prominent during wartime, while general

Case Study: Agriculture and Food Policy



Period	N	Substantive	General	Institutional
Postwar 1920s	4,935	39.39%	40.65%	19.96%
Depression 1930s	5,546	41.80%	31.25%	26.96%
War 1940s	1,118	39.39%	31.25%	29.36%
Postwar 1940s	1,897	39.39%	31.25%	29.36%

Methodological Contribution

WHAT THE WORKFLOW CONTRIBUTES

- ▶ It audits LLM-generated labels rather than accepting them as ground truth.
- ▶ It distinguishes topic content from discourse form.
- ▶ It turns label-cluster mismatch into an interpretable object of analysis.

WHAT SHOULD REMAIN CAUTIOUS

- ▶ LLM labels are provisional semantic references.
- ▶ KMeans captures lexical similarity, not deep semantic meaning.
- ▶ The three discourse types are researcher-defined interpretations.
- ▶ The time trends are descriptive, not causal estimates.

Final Takeaway

The LLM tells us what Parliament was talking about.

KMeans helps us see how Parliament talked about it.

**The value of KMeans is not that it replaces LLM labels,
but that it makes their blind spots analytically visible.**

Thank you

Questions and comments are welcome.